



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B. Tech (CBCS) Degree Examinations
 May 2025

ME1301-1 – INTRODUCTION TO AIRCRAFT DESIGN
 (For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Who succeeded in building & testing first Supersonic jet?**
 A) Russians B) Germans
 C) British D) USA
2. **Spoilers are used to**
 A) Increase lift and Decrease Drag B) Decrease drag
 C) Roll, Decrease lift and as Speed brakes D) Pitch up the aircraft
3. **Empennage refers to**
 A) Landing Gear B) Trailing edge devices
 C) Tail Structure of the aircraft D) Nose Portion of the aircraft
4. **Which of the following is a leading-edge device?**
 A) Slat B) Flap
 C) Aileron D) Winglet
5. **The speed regime in which Coefficient of Drag is highest is**
 A) Subsonic B) Transonic
 C) Supersonic D) Hypersonic
6. **The angle of attack at which highest lift is observed is**
 A) $0^\circ - 5^\circ$ B) $6^\circ - 12^\circ$
 C) $12^\circ - 15^\circ$ D) above 16°
7. **Ribs are used in wings**
 A) As Main strengthening members B) To resist shear torsion loads
 C) To provide the aerofoil shape D) To resist bending
8. **The wing sweepback reduces the effect of**
 A) Induced drag B) Form drag
 C) Wave drag D) Skin friction drag
9. **Aircraft is in a level unaccelerated flight. If the aircraft is flying at a high speed, the angle of attack will be**
 A) medium B) low
 C) high D) moderate
10. **At cruise condition L/D ratio is**
 A) 1.0 B) cant say
 C) maximum D) Maximum
11. **Buffeting refers to**
 A) Fuselage vibration B) Landing gear vibration
 C) Horizontal tail vibration D) Wing vibration

12. In an emergency landing condition, the forward g specified is
 A) 6.0g B) 4.5g
 C) 9g D) -3.5g
13. In a steady banked turn, the lift is
 A) equal to weight B) less than weight
 C) greater than the weight D) double the weight
14. Ratio of climb is proportional to
 A) drag B) Excess of thrust over drag
 C) thrust D) weight of A/C
15. Pilot uses pedals to control
 A) Elevator B) Rudder
 C) Aileron D) Flap
16. In a steady banked turn, the lift is
 A) equal to weight B) less than weight
 C) greater than the weight D) double the weight
- Vapor lock leads to:
 17. A) high pressure in fuel supply B) low pressure in fuel supply
 C) high temperature in fuel supply D) low temperature in fuel supply
- Dead Reckoning Navigation System uses
 18. A) Compass, Time to a destination and Speed of flight to Navigate B) On ground land marks to navigate
 C) Electronic Beamers and Codes D) Radio instruments
- Today's modern Aircraft use the following navigation while flying over large bodies of water
 19. A) VOR B) GPS
 C) DME D) NDB
- The Symbol in a Aeronautical Map represents
 20.  A) NDB B) VOR
 C) DME D) GPS

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With neat sketch explain different structural elements of aircraft wing.	8	L*2	1	1
	b) Briefly explain the following: i. Indicated Air speed ii. Calibrated Air speed iii. Equivalent Air speed iv. True Air speed	8	L2	2	1
2.	a) Briefly explain aircraft development between World War I and World War II.	8	L2	1	1
	b) With the help of necessary sketches explain i. Angle of attack ii. Aspect ratio iii. Taper ratio iv. Chord	8	L2	2	1
3.	a) Explain the different motions of the aircraft with a neat sketch. Also explain about the control surfaces used to achieve these motions.	8	L2	1	1
	b) Explain the different types of flaps.	8	L2	2	1
Unit – II					
4.	a) Briefly explain the power curves.	8	L2	3	1
	b) Briefly explain the correct and incorrect angles of bank.	8	L2	4	1

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|-------|--|---|----|---|---|
| 5. a) | An aero plane weighs 10000N, the drag in normal horizontal flight is 1250N, the center of pressure is 25mm behind the center of gravity and the line of drag is 150mm above the line of thrust. Find the load on the tail plane which is 6m behind the center of gravity will be required to maintain balance in normal horizontal flight. | 8 | L2 | 3 | 1 |
| b) | Briefly explain the following Aerobatics | | | | |
| i. | Inverted loop | 8 | L2 | 4 | 1 |
| ii. | Spin | | | | |
| 6. a) | With neat sketches explain factor affecting lateral stability. | 8 | L2 | 3 | 1 |
| b) | Write a brief note on manoeuvrability and Balanced controls. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Draw a sketch to show the working of an open-centered hydraulic system. | 8 | L2 | 5 | 1 |
| b) | What are the different types of turbines used in aircrafts? With a neat sketch explain the turbofan engine | 8 | L2 | 5 | 1 |
| 8. a) | List the different mechanical aircraft systems. Explain the fuel gauging systems used in aircraft. | 8 | L2 | 5 | 1 |
| b) | Explain the functions of avionics systems. What are the methods of navigation? | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
